⛳ Fuel Golf — Teacher Guide

Orbital mechanics & the Oberth effect, taught by a fuel gauge

WHAT STUDENTS DO

Each mission gives a spacecraft, a goal (circularize, raise an orbit, escape, transfer, moon flyby) and a **par Δv budget**. Real Newtonian gravity is integrated live, and **the planet is Earth at true scale** (μ = 398,600 km³/s², R = 6,371 km): speeds in km/s, Δv in m/s, altitudes in km — every number transfers to real missions. Students time-warp, aim a burn, preview the dotted trajectory, commit. Lowest total Δv wins. Runs offline in any browser — open **index.html** (Chromebook-friendly). The moon is fictional (closer/heavier than Luna) so an assist fits one class.

20-MINUTE SESSION PLAN

- **0–3 min** Project the game (⚙ → Projector mode). Fly Level 1 together: show time-warp, the burn planner, and the dotted preview. Vocabulary: periapsis (low, fast), apoapsis (high, slow), prograde.
- **3–8 min** Students play Levels 1–2 in pairs. Ask them to open the **Physics HUD** and watch “energy gained per 1 Δv” as the ship swings around.
- **8–15 min** The main event: **Level 3, “Escape Artist.”** Par is 791 m/s, but the tank holds 2,057 — escaping Earth from apoapsis costs ≈1,835 m/s, while the same escape from periapsis costs ≈661. Let them discover this. When someone makes par, have them explain *where* they burned and what the HUD showed.
- **15–19 min** Debrief on the projector using a student’s energy-vs-time plot and the **“show the math” panel** below it, which evaluates $\Delta\varepsilon = v \cdot \Delta v + \frac{1}{2}\Delta v^2$ with that student’s actual numbers and checks it against the simulation. The burn is a vertical jump in ε — taller when v was larger, for the *same* Δv.
- **19–20 min** Post the leaderboard. Levels 4–7 (Hohmann, powered flyby, finite thrust) make excellent homework or a second session.

THE PHYSICS — OBERTH EFFECT

An impulsive burn changes speed by Δv along the velocity. Kinetic energy per unit mass goes from $\frac{1}{2}v^2$ to $\frac{1}{2}(v+\Delta v)^2$, so

$$\Delta KE = v \cdot \Delta v + \tfrac{1}{2}\Delta v^2$$

The **v·Δv** term dominates: the same Δv buys energy in proportion to how fast you are already moving. On an orbit, speed peaks at periapsis — so a burn there buys the most orbital energy $\varepsilon = \frac{1}{2}v^2 - \mu/r$. Gravitational potential is unchanged at the instant of the burn; only the KE jump differs. In the game, the HUD’s “energy per unit Δv” readout is exactly v, and students can verify the ≈2.8× escape-cost gap on Level 3 themselves.

Where does the extra energy come from? Rocket fuel’s energy is fixed, but the useful energy delivered to the *ship* is force × distance. At high speed the engine acts over a longer distance per second, so more of the propellant’s energy ends up in the ship and less in the exhaust left behind.

Misconception 1: “The burn is better at periapsis because you’re *closer* to the planet.” Proximity is not the mechanism — **speed** is. $\Delta KE = v \cdot \Delta v + \frac{1}{2}\Delta v^2$ contains v, not r. Being deep in the well matters only because falling there is what made the ship fast. A ship moving equally fast far from the planet would gain just as much energy from the same burn.

Misconception 2: “The Oberth effect is the same as a gravity assist.” An **unpowered gravity assist** (Level 5’s moon) steals energy from the *moon’s orbital motion* — no fuel needed, and the moon slows imperceptibly. An **Oberth burn** spends fuel and merely spends it at the most profitable moment. A “powered flyby” combines both.

Misconception 3: “Δv is like distance — it costs the same anywhere.” Δv *spent* is the same anywhere; the *energy it buys* is not. That is the whole game.

FINITE THRUST (LEVELS 6–7)

Levels 1–5 use impulsive burns; levels 6–7 give the engine limited acceleration (0.055 and 0.013 m/s²), so Δv is delivered over hours of arc (steering losses ignored). Two lessons emerge: a single burn is cheapest **centered on periapsis** — start early, or the late half happens at lower speed (lit at Pe: 919 m/s to escape; centered: 775) — and with the weak engine one long burn costs 1,498 m/s while six ~3 h **perigee kicks** cost 731. This is why real upper stages escape via repeated perigee kicks.

DISCUSSION QUESTIONS

1. Level 3’s energy plot shows a taller jump in ε for the same Δv at periapsis. Which term of $\Delta KE = v \cdot \Delta v + \frac{1}{2}\Delta v^2$ explains this?
2. The HUD says “energy per unit Δv = current speed.” Differentiate $\frac{1}{2}v^2$ to show why.
3. Why does a prograde burn at periapsis raise the *apoapsis*, on the far side of the orbit?
4. In Level 5, where does the escape energy come from when you fly close behind the moon without burning? What slows down in exchange?
5. Real missions (e.g., Parker Solar Probe, Voyager) plan burns at closest approach. Which levels model this, and why does NASA care so much about Δv budgets?
6. Could you escape with a radial-out burn? Try it. Why does pushing “straight up” waste fuel?

CLASSROOM LOGISTICS

- **Leaderboards** persist per level on each machine; reset them between class periods via ⚙ Teacher mode → Reset.
- Pars (m/s): L1 712 · L2 1,107 · L3 791 · L4 1,819 · L5 1,424 · L6 838 · L7 823. Verified minimums: L3 escape ≈ 661 m/s from periapsis vs ≈ 1,835 m/s from apoapsis.
- Keyboard: **Space** pause · **B** burn planner · **X** cut engine · **, .** warp down/up · **H** HUD.